

220B Discussion 4

Problem 1: Admissibility on $\lambda(S)$ in a 2×3 game

Consider a two-person zero-sum game with $\Theta = \{\theta_1, \theta_2\}$, $\mathcal{A} = \{a_1, a_2, a_3\}$, and loss matrix

	a_1	a_2	a_3
θ_1	4	3	0
θ_2	0	3	4

- (a) Compute the three nonrandomized risk points and sketch the risk set $S \subseteq \mathbb{R}^2$.
- (b) Show that a_2 is inadmissible by exhibiting an explicit randomized rule that strictly dominates it. State the corresponding risk point.
- (c) Identify the lower boundary $\lambda(S)$. Find the minimax risk point s_M , the value V , and the minimax randomized strategy δ_M^* .
- (d) Find the maximin prior π_M from the tangent-hyperplane characterization. Verify $L(\theta, \delta_M^*) \leq V \leq L(\pi_M, a)$ for every θ and every $a \in \{a_1, a_2, a_3\}$, noting which inequality is strict and why.

Problem 2: Two losses, two least-favorable priors

Let $X \sim \text{Binomial}(n, p)$ with $p \in [0, 1]$ and action space $\mathcal{A} = [0, 1]$.

Under squared-error loss $L(p, a) = (p - a)^2$, Lecture 8 (Example 17) established that the minimax rule is

$$\delta^{*M}(X) = \frac{X + \sqrt{n}/2}{n + \sqrt{n}},$$

with least-favorable prior $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$.

Now consider the **weighted squared-error loss**

$$\tilde{L}(p, a) = \frac{(a - p)^2}{p(1 - p)}, \quad p \in (0, 1).$$

(a) Compute $R(p, X/n)$ under $\tilde{L}$. Conclude that the MLE X/n is an equalizer rule and state its constant risk.

(b) Show that under $\tilde{L}$, the Bayes action at X against any prior π with finite posterior moments equals $\mathbb{E}_{\pi}[1/(1 - p) \mid X] / \mathbb{E}_{\pi}[1/(p(1 - p)) \mid X]$.

(c) Show that X/n is the Bayes rule under $\tilde{L}$ against the **Uniform** prior $\pi(p) = \mathbf{1}\{p \in (0, 1)\}$, for $X \in \{1, 2, \dots, n - 1\}$.

(d) Apply Theorem 17 to conclude X/n is minimax under $\tilde{L}$. State the least-favorable prior under $\tilde{L}$, and contrast it with the least-favorable prior $\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ under squared-error loss. In one or two sentences: what does this comparison say about the role of the loss function in shaping the minimax solution?